

Integrated Planning of Cyber-Physical Active Distribution System Considering Multidimensional Uncertainties

Hongjun Gao¹, Senior Member, IEEE, Xiaodong Lyu, Shuaijia He, Lingfeng Wang, Senior Member, IEEE, Cheng Wang², Member, IEEE, and Junyong Liu, Member, IEEE

Abstract—The integration of cyber technologies is transforming the power distribution network into a cyber-physical active distribution system (CPADS), and brings more flexibility to its operation. However, various uncertainties from the cyber space and physical space are emerging and coupled with each other, which results in significant challenges for the distribution network planning and operation. Thus, a bilevel integrated planning model accounting for multidimensional uncertainties from both the cyber space and physical space is proposed for the CPADS in this paper. With the goal of minimizing the annual investment and operation costs, the model comprehensively considers the location and selection of active-management elements (e.g., electrical storage systems, capacitor banks, and switches), investment constraints, power flow constraints, and operation constraints for all the active-management elements, etc. Moreover, based on the typical cyber system failure scenario represented by information link failure (ILF), a novel N - k uncertainty set considering the failure probability distribution is proposed, while the uncertainties of renewable energy and load demand are characterized by box-like uncertainty sets. Finally, the proposed planning model is recast into a multi-stage robust optimization model which can be divided into a master problem and two-step subproblems (SP-1 and SP-2) to be solved by the column and constraint generation (CCG) algorithm. Case studies and comparative analysis on the PG 69 distribution system are performed to verify the effectiveness of the proposed method.

Index Terms—Smart distribution network planning and operation, cyber-physical active distribution system (CPADS), N - k uncertainty set, information link failure (ILF).

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Hongjun Gao, Xiaodong Lyu, Shuaijia He, and Junyong Liu are with the College of Electrical Engineering, Sichuan University, Chengdu 610065, China (e-mail: gaohongjun@scu.edu.cn; 1825322276@qq.com).

Lingfeng Wang is with the Department of Electrical Engineering and Computer Science, University of Wisconsin–Milwaukee, Milwaukee, WI 53211 USA.

Cheng Wang is with the State Key Laboratory of Alternate Electrical Power System with Renewable Energy Sources, North China Electric Power University, Beijing 102206, China.

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NOMENCLATURE

Indices

i, j, k
 ij, jk
 l
 t

Index of buses.
Index of branches.
Index of candidate unit types.
Index of time periods.

Sets

$\Omega^{\text{ESS}}/\Omega^{\text{Switch}}$
 $\delta(j)/\pi(j)$
 $\mathcal{V}$
 $\xi, \xi^{\text{WTG}}/\xi^{\text{PVG}}/\xi^{\text{Load}}$

Set of unit types of electrical storage system (ESS)/switch.
Set of buses whose parent/child is bus j .
Set of uncertainty from the cyber space.
Set of uncertainty from the physical space, sets of uncertainty parameters of WTG/PVG/Load.
Set of system buses/load buses/photovoltaic generator (PVG) buses/wind turbine generator (WTG) buses/substation bus.
Set of candidate buses for installing an ESS/capacitor bank (CB).
Set of system branches, candidate branches for installing a switch.

$B/B^{\text{Load}}/B^{\text{PVG}}/B^{\text{WTG}}/B^{\text{Sub}}$

$B^{\text{ESS}}/B^{\text{CB}}$

E, B^{Switch}

Parameters

$\alpha_j^{\text{charge}}/\alpha_j^{\text{discharge}}$
 Δt
 β
 $\psi_{j,t}^{\text{ESS,PRE}}/\psi_{j,t}^{\text{CB,PRE}}/$
 $\psi_{j,t}^{\text{Switch,PRE}}/$
 $c_l^{\text{ESS}}/c_l^{\text{CB}}/c_l^{\text{Switch}}$
 $c_l^{\text{FRTU,ESS}}/c_l^{\text{FRTU,Switch}}/c_l^{\text{FRTU,CB}}$

Charging/discharging efficiency coefficient of ESS.
Time interval.
Operation equivalent factor.
Probability for information link failure in the FRTU of ESS/CB/switch.
Price for type- l ESS/CB/type- l switch.
Price for feeder remote terminal unit (FRTU) of type- l ESS/type- l switch/CB.
Price for power loss/wind power curtailment/solar

$c_t^{\text{Loss}}/c_t^{\text{WTG}}/c_t^{\text{PVG}}/c_t^{\text{ENS}}$

$E_l^{\text{ESS}}/\bar{E}_l^{\text{ESS}}$	power curtailment/energy not supplied (ENS).	$P_{j,t}^{\text{charge}}/P_{j,t}^{\text{discharge}}$	ESS charging/discharging power.
$I_{ij}/\bar{I}_{ij}$	Lower/upper bound of capacity for type- l ESS.	$P_{j,t}^{\text{ENS}}/Q_{j,t}^{\text{ENS}}$	Active/reactive load power not supplied.
M^{CB}	Lower/upper bound of current value.	$P_{j,t}^{\text{PVG}}$	Active solar power.
$N^{\text{Bus}}, N^{\text{Sub}}$	Maximum banks of CB to be installed.	$P_{j,t}^{\text{Sub}}/Q_{j,t}^{\text{Sub}}$	Main-grid (substation) active/reactive power.
$\bar{N}^{\text{CB}}$	Total number of buses, total number of substation buses.	$P_{j,t}^{\text{WTG}}/Q_{j,t}^{\text{WTG}}$	Active/reactive wind power.
$\bar{N}_l^{\text{ESS}}/\bar{N}_l^{\text{Switch}}$	Maximum number of regulations for CB.	$P_{ij,t}^*$	Auxiliary active power.
$P_l^{\text{ESS}}/\bar{P}_l^{\text{ESS}}$	Maximum number of regulations for type- l ESS/type- l switch.	$\bar{V}_{j,t}$	Square of voltage.
$P_{j,t}^{\text{Load,PRE}}/Q_{j,t}^{\text{Load,PRE}}$	Lower/upper bound of power of type- l ESS.	$x^{\text{Inv}}/x^{\text{Ope}}$	Planning/operation decision variable.
$\bar{P}_j^{\text{Load,PRE}}/\bar{P}_j^{\text{Load,PRE}}$	Forecasted active/reactive power.	x_j^{CB}	Binary variable; it equals to 1 if installing CB at bus j .
$P_{j,t}^{\text{PVG,PRE}}$	Lower/upper bound of forecasted active power.	$x_{j,l}^{\text{ESS}}$	Binary variable; it equals to 1, if installing type- l ESS at bus j .
$\bar{P}_{j,t}^{\text{PVG,PRE}}/\bar{P}_{j,t}^{\text{PVG,PRE}}$	Forecasted output of solar power.	$x_{ij,l}^{\text{Switch}}$	Binary variable; it equals to 1, if installing type- l switch at branch ij .
$\bar{P}_j^{\text{Sub}}/\bar{P}_j^{\text{Sub}}$	Lower/upper bound of main-grid (substation) active power.	$y_{j,t}^{\text{CB}}$	Integer variable; bank quantity of CB put into operation.
$P_{j,t}^{\text{WTG,PRE}}/\bar{P}_{j,t}^{\text{WTG,PRE}}$	Lower/upper bound of forecasted output of wind power.	z_j^{CB}	Integer variable; bank quantity of CB to be installed.
$P_{ij,t}^*/\bar{P}_{ij,t}^*$	Lower/upper bound of auxiliary active power flow.		
$Q_{j,t}^{\text{CB}}, Q_j^{\text{CB,step}}$	Reactive power of CB, Reactive power of each CB unit.		
$\bar{Q}_j^{\text{Load,PRE}}/\bar{Q}_j^{\text{Load,PRE}}$	Lower/upper bound of forecasted reactive power.		
$\bar{Q}_j^{\text{Sub}}/\bar{Q}_j^{\text{Sub}}$	Lower/upper bound of main-grid (substation) reactive power.		
$\bar{Q}_{j,t}^{\text{WTG}}/\bar{Q}_{j,t}^{\text{WTG}}$	Lower/upper bound of reactive wind power.		
$V_j/\bar{V}_j$	Lower/upper bound of voltage magnitude.		
Variables			
$\delta_{j,t}^{\text{CB,IN}}/\delta_{j,t}^{\text{CB,DE}}$	Binary variable; on/off status of CB.		
$\delta_{ij,t}^{\text{Switch,IN}}/\delta_{ij,t}^{\text{Switch,DE}}$	Binary variable; on/off status of switch.		
$\gamma_{j,t}^{\text{ESS}}/\gamma_{j,t}^{\text{CB}}/\gamma_{ij,t}^{\text{Switch}}$	Binary variable; it equals to 0 if an information link failure occurs for the FRTU of ESS/CB/switch.		
$\mu_{j,t}^{\text{charge}}/\mu_{j,t}^{\text{discharge}}$	Binary variable; Charging/discharging status of ESS.		
$\sigma_{ij,t}$	Binary variable; it equals to 1 if the switch is closed.		
$\tilde{I}_{ij,t}$	Square of current.		
$P_{ij,t}/Q_{ij,t}$	Active/reactive power.		

I. INTRODUCTION

THE FAST development of smart distribution networks accelerates the coupling between its cyber system and physical system, making the information flow and energy flow more interactive. As a result, the traditional distribution network is gradually evolving into a cyber-physical active distribution system (CPADS) [1], [2]. While the wide deployment and application of cyber systems can better facilitate the flexible control of distribution networks and integration of renewable energy, they may inevitably bring some uncertainties to the cyber space of the power grid. Further, the uncertainties from the cyber space may be coupled with the original uncertainties already existing in the physical space, bringing great challenges to the planning and operation of modern distribution networks [3], [4].

There has been limited research on CPADS so far, and the research on active distribution systems (or active distribution networks, ADNs) planning has been relatively more extensive. In [5], a short-term planning framework aiming at minimizing the energy loss of photovoltaic systems was proposed. Reference [6] proposed a scenario matrix based on a heuristic matching technique that could capture the stochastic characteristics and correlation between the distributed generation (DG) and electricity demand. Then, a scenario-based ADN planning model was established. Based on the scenario matrix in [6], Ehsan and Yang [7] proposed a scenario-based stochastic ADN planning model to maximize the net annual profit. Specifically, the location, as well as the selection of DGs and electrical storage systems (ESSs), were considered. In [8], a bilevel stochastic framework for ADN planning was proposed, which fully considered the system framework, renewable energy resources, and distribution static compensators. In [9], a bilevel coordinated planning model of converter-based DG units and soft open points was established in an unbalanced distribution network. Li *et al.* [10] proposed a multi-objective

and multi-level model involving DGs and ESSs, where the upper level and the middle level modeled the planning problems from different perspectives of multi-stakeholders, while the lower level modeled the operation of ESSs. All the above studies took into account the uncertainties from the physical space (such as renewable energy and load demand). However, the uncertainties from the cyber space were neglected.

Recently, the reliability of cyber-physical system (CPS) has drawn wide attentions from the researchers. In [11], a distribution network CPS correlation matrix based on the topological structure was established to reflect the power-information coupling characteristics. In [12], a reliability modeling and evaluation method for CPS was proposed based on a composite Markov model to couple physical characteristics and information flow performance. To cope with the impact of cyber network congestion, a CPS robust routing model with a priority mechanism considering cyber-physical disturbances was proposed based on robust optimization in [13]. References [14]–[16] modeled the coupling relationship of the microgrid CPS, then quantitatively evaluated the impact of disturbances from the cyber space on the reliability of microgrid using a sequential Monte Carlo simulation method. In [17], the physical response of ADN to malicious cyber-attacks was elaborately investigated, where the potential effect and damage of cyber-attacks on ADN were assessed. The above studies focused on the cyber topology analysis and interface modeling, but the modeling of the physical system could be further extended. The existing models may not be sufficiently granular to describe the complex interactions between the discrete states of the cyber space and the continuous dynamics of the physical space, as well as to model the increased risk incurred by the cyber-physical interplay. Thus, it is meaningful to construct a comprehensive ADS spatial-temporal operation model considering the strong coupling of CPS.

Meanwhile, robust optimization is an effective mathematical method for coping with uncertainties. Uncertainty sets are usually used to characterize uncertainties in robust optimization problems, such as the box-like uncertainty set, polyhedral uncertainty set, and N - k uncertainty set. References [18]–[21] used box-like uncertainty sets to characterize the uncertainties of renewable energy and load demand, while [22]–[23] used polyhedral uncertainty sets. And all of them constructed a two-stage robust optimization model, which was solved by the column and constraint generation (CCG) algorithm. Although the robust optimization results based on box-like uncertainty set is slightly more conservative than those based on polyhedral uncertainty sets, box-like uncertainty set is commonly used especially when applied into the practical application due to its simple structure to easily perform dualization. This makes the robust optimization model more feasible and efficient to be solved. In addition, N - k uncertainty sets are usually adopted to characterize low-probability high-impact failures. In [24], a robust optimization model for unit commitment considering the N - k security criterion was constructed. In [25], a robust scheduling model considering N - k contingencies for ADN against windstorms was proposed. In [26], a robust optimization model for ADN configuration considering the distribution lines' N - k contingencies was established.

Li *et al.* [27] formulated a robust unit commitment model considering N - k contingencies for enhancing the resilience of the integrated power distribution and natural gas system. However, the optimization results in [24]–[27] generally tend to be overly conservative, since the occurrence probability of the worst-case scenarios is usually quite low. Therefore, it is necessary to take into account the distribution of failure probabilities in the construction of N - k uncertainty sets.

To address the aforementioned issues, an integrated bilevel planning method for CPADS is proposed in this paper considering multidimensional uncertainties from both cyber and physical spaces. Firstly, an integrated CPADS planning-operation framework is proposed. A model aiming at minimizing the annual comprehensive cost of investment and operation is established considering the selection and location of active-management elements, such as ESSs, capacitor banks (CBs) and switches. The constraints of this model comprehensively include the investment constraints, power flow constraints, system security constraints, and operation constraints on active-management elements, etc. Specifically, the strong coupling relationship of cyber-physical systems is embodied in the detailed technical constraints of active-management elements. Secondly, multidimensional uncertainty sets from the cyber space and physical space are constructed. Based on the typical cyber system failure scenario represented by the information link failures (ILFs), an N - k uncertainty set considering the failure probability distribution is proposed for the cyber space. Meanwhile, the uncertainties of wind power, photovoltaic power and load demand in the physical space are described as box-like uncertainty sets. Thirdly, the proposed multi-stage robust optimization model is divided into a master problem (MP) and two-step subproblems (SP-1 and SP-2). Specifically, SP-1 is formulated to obtain the worst-case scenario in the physical space, while SP-2 is formulated to obtain the worst-case scenario in the cyber space. The whole model is solved by the CCG algorithm iteratively. Finally, case studies and comparative analysis are carried out based on the PG 69 distribution system, and the performance of the proposed method is well demonstrated.

The rest of this paper is organized as follows. In Section II, the CPDAS and the ILF are analyzed. In Section III, the bilevel CPADS planning model and multidimensional uncertainties from cyber space and physical space are presented. In Section IV, the formulated multi-stage robust optimization model and the solution methodology are described in detail. In Section V, numerical results based on the PG 69 distribution system are provided. Finally, conclusions are drawn in Section VI.

II. CYBER-PHYSICAL ACTIVE DISTRIBUTION SYSTEM AND INFORMATION LINK FAILURE

A representative structure of a CPADS can be divided into a cyber system and a physical system, as shown in Fig. 1. The cyber system is composed of the application layer, communication layer and interface layer. The application layer may consist of the distribution automation center, supervisory control and data acquisition (SCADA) server and management

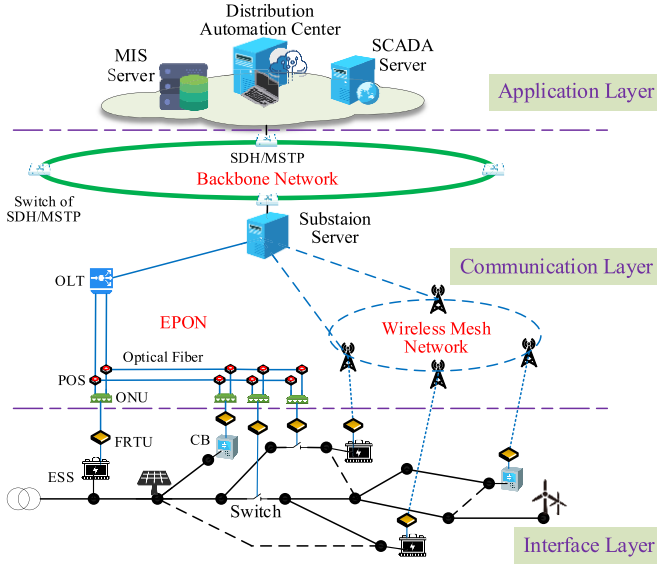


Fig. 1. The representative structure of CPADS.

information system (MIS) server. The communication layer is supported by several network modes including wireless mesh network [28], double-chain ethernet passive optical network (EPON) [1] and backbone network based on the synchronous digital hierarchy (SDH) or multi-service transform platform (MSTP). The double-chain EPON is realized with optical fibers, optical line terminals (OLTs), passive optical splitters (POSS), and optical network units (ONUs). Especially, the backbone network is connected with the application layer and various substation servers. With the help of intelligent terminals, such as the feeder remote terminal units (FRTUs), the interface layer plays a significant role in keeping the real-time communication among various active-management elements (e.g., ESS, CB and switch).

The cyber system plays a data upload and control signal delivery role, which is obviously essential for better operating the physical distribution system. All the active-management element are equipped with FRTUs, which are responsible for not only measuring and uploading data-information such as voltage, current, power, and device working status, but also receiving and executing remote control instructions. In the practical application, the data-information gathered by FRTUs should be firstly identified and de-redundant, and then uploaded to the application layer by the communication layer. Afterwards, with the assistance of the MIS server and the SCADA server, the distribution automation center makes the optimal operation strategy after analyzing all collected data-information. Finally, the control instructions are delivered to each FRTU via the communication layer.

In addition, the information link failure is a significant issue when studying CPADS, which defined as the information cannot be transmitted through the information link [2]. It is mainly caused by insufficient information transmission power, the channel environment deterioration, traffic congestion, protocol errors, and external force damage, etc. The information link refers to the whole transmission channel

from the application layer to the interface layer. As shown in Fig. 1, the application layer server of CPADS is typically well protected and maintained. Meanwhile, the ring structure backbone network, the double-chain EPON, and the wireless mesh network in the communication layer can quickly enable the standby of a channel to ensure communication reliability if a certain channel in the information link is interrupted. Thus, the occurrence probability of ILFs in the application layer and communication layer is pretty low. Besides, this study focuses on the impacts of cyber system failure on the physical system. It is obvious that the control instructions from the application layer can be responded immediately by the devices in the physical system only when the information link is available. That is to say, the availability of those physical active management devices (e.g., ESS, CB) is the key to describe cyber (information) link failure under the interaction between the cyber system and the physical system. Also, the availability of those physical devices is equivalently expressed by the FRTU working status. In general, the occurrence of ILFs is a threat to the active-control over the physical system operation. The multidimensional uncertainties including ILF uncertainty and the original physical uncertainty are inevitable to result in significant challenges for the planning and operation of distribution networks.

III. INTEGRATED CPADS PLANNING PROBLEM FORMULATION

A. Framework of the Bilevel Planning Model

The planning model in this study considers both active-management elements in the physical layer and FRTUs in the cyber layer. The overall framework of the bilevel planning model can be represented in following equation (1). The upper level is the investment level while the lower level is the operation level. As can be seen from (1), the objective is to minimize the total cost C^{Total} , which includes the investment cost C^{Inv} and operation cost C^{Ope} . $G(\cdot)$ and $H(\cdot)$ represent the inequality constraints and equality constraints of the upper investment level, respectively. $g(\cdot)$ and $h(\cdot)$ represent the inequality constraints and equality constraints in the lower operation level, respectively. It should be noted that the upper level and lower level are coupled with each other. Traditionally, the optimized planning results (active-management elements) are delivered to the lower level for operation simulation. Then, the operation cost results at the lower level are returned to upper level to influence the planning solution. Generally, such a bilevel model is solved iteratively through the interaction between the upper and lower level.

$$\begin{cases} \min_{x^{\text{Inv}}} C^{\text{Total}}(x^{\text{Inv}}, x^{\text{Ope}}) = C^{\text{Inv}} + C^{\text{Ope}} \\ \text{s.t.} \begin{cases} G(x^{\text{Inv}}) \leq 0 \\ H(x^{\text{Inv}}) = 0 \end{cases} \\ \max_{\{\xi\}|\{\gamma\}} \left(\min_{x^{\text{Ope}}} f(x^{\text{Inv}}, x^{\text{Ope}}) \right) = C^{\text{Ope}} \\ \text{s.t.} \begin{cases} g(x^{\text{Inv}}, x^{\text{Ope}}, \{\xi, \gamma\}) \leq 0 \\ h(x^{\text{Inv}}, x^{\text{Ope}}, \{\xi, \gamma\}) = 0 \end{cases} \end{cases} \quad (1)$$

For the active-management elements to be planned, three kinds of decision variables are introduced in this paper, i.e.,

planning decision variables, scheduling decision variables and information link availability variables. The planning decision variables indicate whether to plan a certain unit-type/number of devices at the candidate locations. The scheduling decision variables characterize the control signals/instructions from the cyber system. The information link availability variables represent whether the information link can work normally, i.e., whether corresponding active-management devices in the physical system can be successfully controlled by the cyber system.

B. Objective Functions

1) *Investment Cost*: The investment cost includes the investment cost of active-management elements (ESSs, CBs and switches) and corresponding FRTUs investment cost, as shown in (2). It is noteworthy that the proposed model in this study just focuses on the static (single-stage) planning problem with both physical and cyber devices considered instead of multi-stage planning. Thus, all the cost terms are discounted to that in the current year.

$$C^{\text{Inv}} = \frac{r(1+r)^{TL_{\text{ESS}}}}{(1+r)^{TL_{\text{ESS}}} - 1} \cdot \sum_{j \in B^{\text{ESS}}} \sum_{l \in \Omega^{\text{ESS}}} (c_l^{\text{ESS}, \text{ESS}} x_{j,l}^{\text{ESS}} + c_l^{\text{FRTU}, \text{ESS}} x_{j,l}^{\text{ESS}}) \\ + \frac{r(1+r)^{TL_{\text{CB}}}}{(1+r)^{TL_{\text{CB}}} - 1} \cdot \sum_{j \in B^{\text{CB}}} (c_j^{\text{CB}, \text{CB}} x_j^{\text{CB}} + c_j^{\text{FRTU}, \text{CB}} x_j^{\text{CB}}) \\ + \frac{r(1+r)^{TL_{\text{Switch}}}}{(1+r)^{TL_{\text{Switch}}} - 1} \cdot \sum_{ij \in E^{\text{Switch}}} \sum_{l \in \Omega^{\text{Switch}}} (c_l^{\text{Switch}, \text{Switch}} x_{ij,l}^{\text{Switch}} \\ + c_l^{\text{FRTU}, \text{Switch}} x_{ij,l}^{\text{Switch}}) \quad (2)$$

In (2), those three terms represent the investment costs of ESS, CB and switch, respectively.

2) *Operation Cost*: The operation cost in (3) includes the cost of network loss, penalty cost of wind power curtailment, penalty cost of solar power curtailment, and penalty cost of the energy not supplied (ENS).

$$C^{\text{Ope}} = 365\beta \cdot \sum_{t \in T} \Delta t \left[\sum_{ij \in E} c_t^{\text{Loss}} \tilde{I}_{ij,t} r_{ij} \right. \\ + \sum_{j \in B^{\text{WTG}}} c_t^{\text{WTG}} (P_{j,t}^{\text{WTG}, \text{PRE}} - P_{j,t}^{\text{WTG}}) \\ + \sum_{j \in B^{\text{PVG}}} c_t^{\text{PVG}} (P_{j,t}^{\text{PVG}, \text{PRE}} - P_{j,t}^{\text{PVG}}) \\ \left. + \sum_{j \in B^{\text{Load}}} c_t^{\text{ENS}} P_{j,t}^{\text{ENS}} \right] \quad (3)$$

In (3), we just consider a single year (365 days) operation simulation because our proposed model is a static (single-stage) planning problem.

C. Constraints

1) *Investment Constraints*: The number of installed devices is restricted as follows:

$$\begin{cases} 0 \leq \sum_{l \in \Omega^{\text{ESS}}} x_{j,l}^{\text{ESS}} \leq 1, & \forall j \in B^{\text{ESS}} \\ 0 \leq z_j^{\text{CB}} \leq M^{\text{CB}}, & \forall j \in B^{\text{CB}} \\ 0 \leq \sum_{l \in \Omega^{\text{Switch}}} x_{ij,l}^{\text{Switch}} \leq 1, & \forall ij \in E^{\text{Switch}} \end{cases} \quad (4)$$

In (4), the first/third constraint stipulates that only one optional type of ESS/switch can be installed at each candidate ESS bus/switch branch. The second constraint concerns the maximum number of devices that can be installed is M^{CB} at each candidate CB-bus.

2) Operation Constraints:

a) *Power flow constraints*: The distflow model is widely adopted to describe the power flow in radial distribution networks, as shown in (5)-(7).

$$\begin{cases} \sum_{k \in \delta(j)} P_{jk,t} - \sum_{i \in \pi(j)} (P_{ij,t} - \tilde{I}_{ij,t} r_{ij}) = \sum_{j \in B^{\text{Sub}}} P_{j,t}^{\text{Sub}} \\ + \sum_{j \in B^{\text{WTG}}} P_{j,t}^{\text{WTG}} + \sum_{j \in B^{\text{PVG}}} P_{j,t}^{\text{PVG}} + \sum_{j \in B^{\text{ESS}}} P_{j,t}^{\text{discharge}} \\ - \sum_{j \in B^{\text{ESS}}} P_{j,t}^{\text{charge}} - \sum_{j \in B^{\text{Load}}} (P_{j,t}^{\text{Load}, \text{PRE}} - P_{j,t}^{\text{ENS}}) \\ \sum_{k \in \delta(j)} Q_{jk,t} - \sum_{i \in \pi(j)} (Q_{ij,t} - \tilde{I}_{ij,t} x_{ij}) = \sum_{j \in B^{\text{Sub}}} Q_{j,t}^{\text{Sub}} \\ + \sum_{j \in B^{\text{WTG}}} Q_{j,t}^{\text{WTG}} + \sum_{j \in B^{\text{CB}}} Q_{j,t}^{\text{CB}} \\ - \sum_{j \in B^{\text{Load}}} (Q_{j,t}^{\text{Load}, \text{PRE}} - Q_{j,t}^{\text{ENS}}) \end{cases} \quad \forall t, \forall j \in B, \forall ij \in E \quad (5)$$

$$\begin{cases} \tilde{V}_{i,t} - 2(P_{jk,t} r_{ij} + Q_{jk,t} x_{ij}) + \tilde{I}_{ij,t} (r_{ij}^2 + x_{ij}^2) \\ \geq \tilde{V}_{j,t} + M \cdot (\sigma_{ij,t} - 1) \\ \tilde{V}_{i,t} - 2(P_{jk,t} r_{ij} + Q_{jk,t} x_{ij}) + \tilde{I}_{ij,t} (r_{ij}^2 + x_{ij}^2) \\ \leq \tilde{V}_{j,t} + M \cdot (1 - \sigma_{ij,t}) \\ \tilde{V}_{i,t} - 2(P_{jk,t} r_{ij} + Q_{jk,t} x_{ij}) + \tilde{I}_{ij,t} (r_{ij}^2 + x_{ij}^2) \\ = \tilde{V}_{j,t}, \quad \forall ij \in E \setminus E^{\text{Switch}} \end{cases} \quad \forall t, \forall j \in B \quad (6)$$

$$\begin{cases} 2P_{ij,t} \\ 2Q_{ij,t} \\ \tilde{I}_{ij,t} - \tilde{V}_{i,t} \end{cases} \leq \tilde{I}_{ij,t} + \tilde{V}_{i,t}, \quad \forall t, \forall ij \in E. \quad (7)$$

b) *Security constraints*: The square values of voltage and current are constrained by (8) to guarantee the system security.

$$\begin{cases} V_j^2 \leq \tilde{V}_{j,t} \leq \bar{V}_j^2, & \forall j \in B \\ \tilde{I}_{ij}^2 \leq \tilde{I}_{ij,t} \leq \bar{I}_{ij}^2, & \forall ij \in E \setminus E^{\text{Switch}} \\ \sigma_{ij,t} \tilde{I}_{ij}^2 \leq \tilde{I}_{ij,t} \leq \sigma_{ij,t} \bar{I}_{ij}^2, & \forall ij \in E^{\text{Switch}} \end{cases} \quad \forall t. \quad (8)$$

c) ESS operation constraints:

$$\begin{cases} \mu_{j,t}^{\text{discharge}} + \mu_{j,t}^{\text{charge}} \leq 1 \\ \sum_{t \in T} (\mu_{j,t}^{\text{discharge}} + \mu_{j,t}^{\text{charge}}) \leq \sum_{l \in \Omega^{\text{ESS}}} \bar{N}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \end{cases} \quad \forall t, \forall j \in B^{\text{ESS}} \quad (9)$$

$$\begin{cases} P_{j,t}^{\text{discharge}} \geq \mu_{j,t}^{\text{discharge}} \gamma_{j,t}^{\text{ESS}} \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \\ P_{j,t}^{\text{discharge}} \leq \mu_{j,t}^{\text{discharge}} \gamma_{j,t}^{\text{ESS}} \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \\ P_{j,t}^{\text{charge}} \geq \mu_{j,t}^{\text{charge}} \gamma_{j,t}^{\text{ESS}} \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \\ P_{j,t}^{\text{charge}} \leq \mu_{j,t}^{\text{charge}} \gamma_{j,t}^{\text{ESS}} \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \end{cases} \quad \forall t, \forall j \in B^{\text{ESS}} \quad (10)$$

$$\begin{cases} \sum_{t \in T} (\alpha_j^{\text{charge}} P_{j,t}^{\text{charge}} - \alpha_j^{\text{discharge}} P_{j,t}^{\text{discharge}}) \cdot \Delta t = 0 \\ E_{j,t}^{\text{ESS}} = E_{j,t-1}^{\text{ESS}} + (\alpha_j^{\text{charge}} P_{j,t}^{\text{charge}} - \alpha_j^{\text{discharge}} P_{j,t}^{\text{discharge}}) \cdot \Delta t \\ \sum_{l \in \Omega^{\text{ESS}}} E_{j,l}^{\text{ESS}} x_{j,l}^{\text{ESS}} \leq E_{j,t}^{\text{ESS}} \leq \sum_{l \in \Omega^{\text{ESS}}} \bar{E}_l^{\text{ESS}} x_{j,l}^{\text{ESS}} \end{cases} \quad \forall t, \forall j \in B^{\text{ESS}} \quad (11)$$

Expression (9) denotes the status-change constraints of charging/discharging and limitations of ESS operation. Further, both the charging power $P_{j,t}^{\text{charge}}$ and discharging power

$P_{j,t}^{\text{discharge}}$ should be limited within appropriate ranges as shown in (10). It can be observed from (10) that the charging/ discharging power range of ESSs is determined not only by the planning decision variable ($x_{j,l}^{\text{ESS}}$), but also by the scheduling decision variables ($\mu_{j,t}^{\text{charge}}, \mu_{j,t}^{\text{discharge}}$) and information link availability variable ($\gamma_{j,t}^{\text{ESS}}$). In (11), the first constraint guarantees that the ESS's energy storage at the beginning and at the end of an operation period is equal. The second line represents the energy storage state constraint. The third constraint dictates that the ESS's capacity should be limited between the upper and lower bounds.

d) *CB operation constraints:*

$$\begin{cases} Q_{j,t}^{\text{CB}} = y_{j,t}^{\text{CB}} Q_j^{\text{CB,step}} \\ 0 \leq y_{j,t}^{\text{CB}} \leq z_j^{\text{CB}} x_j^{\text{CB}} \end{cases}, \quad \forall t, \forall j \in B^{\text{CB}} \quad (12)$$

$$\begin{cases} \delta_{j,t}^{\text{CB,IN}} + \delta_{j,t}^{\text{CB,DE}} \leq 1 \\ \sum_{t \in T} (\delta_{j,t}^{\text{CB,IN}} + \delta_{j,t}^{\text{CB,DE}}) \leq \bar{N}^{\text{CB}} x_j^{\text{CB}} \end{cases}, \quad \forall t, \forall j \in B^{\text{CB}} \quad (13)$$

$$\begin{cases} y_{j,t}^{\text{CB}} - y_{j,t-1}^{\text{CB}} \leq \delta_{j,t}^{\text{CB,IN}} \gamma_{j,t}^{\text{CB}} x_j^{\text{CB}} - \delta_{j,t}^{\text{CB,DE}} \gamma_{j,t}^{\text{CB}} x_j^{\text{CB}} \\ y_{j,t}^{\text{CB}} - y_{j,t-1}^{\text{CB}} \geq \delta_{j,t}^{\text{CB,IN}} \gamma_{j,t}^{\text{CB}} x_j^{\text{CB}} - \delta_{j,t}^{\text{CB,DE}} \gamma_{j,t}^{\text{CB}} x_j^{\text{CB}} \end{cases}, \quad \forall t, \forall j \in B^{\text{CB}} \quad (14)$$

Expression (12) represents the discrete reactive power constraints of CBs. Expression (13) represents the status- transition limit between on and off, and the operation limit of CB. Expression (14) restricts the regulation range of CBs. It is obvious that the regulation of CBs is determined not only by the planning decision variables ($x_j^{\text{CB}}, z_j^{\text{CB}}$), but also by the scheduling decision variables ($\delta_{j,t}^{\text{CB,IN}}, \delta_{j,t}^{\text{CB,DE}}$) and information link availability variable ($\gamma_{j,t}^{\text{CB}}$).

e) *Switch operation constraints:*

$$\begin{cases} \delta_{ij,t}^{\text{Switch,IN}} + \delta_{ij,t}^{\text{Switch,DE}} \leq 1 \\ \sum_{t \in T} (\delta_{ij,t}^{\text{Switch,IN}} + \delta_{ij,t}^{\text{Switch,DE}}) \leq \sum_{l \in \Omega^{\text{Switch}}} \bar{N}_l^{\text{Switch}} x_{ij,l}^{\text{Switch}} \end{cases}, \quad \forall t, \forall ij \in E^{\text{Switch}} \quad (15)$$

$$\begin{cases} \left(\delta_{ij,t}^{\text{Switch,IN}} \gamma_{ij,t}^{\text{Switch}} \sum_{l \in \Omega^{\text{Switch}}} x_{ij,l}^{\text{Switch}} \right) - \left(\delta_{ij,t}^{\text{Switch,DE}} \gamma_{ij,t}^{\text{Switch}} \sum_{l \in \Omega^{\text{Switch}}} x_{ij,l}^{\text{Switch}} \right) \geq \sigma_{ij,t} - \sigma_{ij,t-1} \\ \left(\delta_{ij,t}^{\text{Switch,IN}} \gamma_{ij,t}^{\text{Switch}} \sum_{l \in \Omega^{\text{Switch}}} x_{ij,l}^{\text{Switch}} \right) - \left(\delta_{ij,t}^{\text{Switch,DE}} \gamma_{ij,t}^{\text{Switch}} \sum_{l \in \Omega^{\text{Switch}}} x_{ij,l}^{\text{Switch}} \right) \leq \sigma_{ij,t} - \sigma_{ij,t-1} \end{cases}, \quad \forall t, \forall ij \in E^{\text{Switch}} \quad (16)$$

Expression (15) denotes the on/off status constraint and the operation frequency constraint of switch. Expression (16) indicates the actual switch status constraint. Obviously, the actual switch status depends not only on the planning decision variable ($x_{ij,l}^{\text{Switch}}$), but also on the scheduling decision variables ($\delta_{ij,t}^{\text{Switch,IN}}, \delta_{ij,t}^{\text{Switch,DE}}$) and information link availability variable ($\gamma_{ij,t}^{\text{Switch}}$).

f) *Radiation and connectivity constraints:*

$$\sum_{ij \in E \setminus E^{\text{Switch}}} 1 + \sum_{ij \in E^{\text{Switch}}} \sigma_{ij,t} = N^{\text{Bus}} - N^{\text{Sub}}, \quad \forall t \quad (17)$$

$$\begin{cases} P_{j,t}^* = \sum_{k \in \delta(j)} P_{jk,t}^* - \sum_{i \in \pi(j)} P_{ij,t}^* = \partial, \quad \forall j \in B \\ \sigma_{ij,t} P_{ij,t}^* \leq P_{ij,t}^* \leq \sigma_{ij,t} \bar{P}_{ij,t}^*, \quad \forall ij \in E^{\text{Switch}} \\ P_{ij,t}^* \leq \bar{P}_{ij,t}^*, \quad \forall ij \in E \setminus E^{\text{Switch}} \end{cases}, \quad \forall t \quad (18)$$

Expression (17) is the radial operation constraint of network reconfiguration. The auxiliary basic linear power balance equations (18) without any DG integration to the distribution system are constructed to avoid the existence of a loop or an island. Specifically, equations (18) are achieved by introducing additional variables (e.g., $P_{jk,t}^*$ and $P_{ij,t}^*$) and making an assumption that a small load power value ∂ exists in each load bus.

g) *Substation power constraints:* The active power and reactive power flowing through a substation are restricted by the substation capacity, as represented in (19).

$$\begin{cases} P_j^{\text{Sub}} \leq P_{j,t}^{\text{Sub}} \leq \bar{P}_j^{\text{Sub}} \\ Q_j^{\text{Sub}} \leq Q_{j,t}^{\text{Sub}} \leq \bar{Q}_j^{\text{Sub}} \end{cases}, \quad \forall t, \forall j \in B^{\text{Sub}}. \quad (19)$$

h) *DG power constraints:* The active power and reactive power contributed by DGs to the distribution network is restricted by the DG capacities, as shown in (20).

$$\begin{cases} 0 \leq P_{j,t}^{\text{PVG}} \leq P_{j,t}^{\text{PVG,PRE}}, \quad \forall j \in B^{\text{PVG}} \\ 0 \leq P_{j,t}^{\text{WTG}} \leq P_{j,t}^{\text{WTG,PRE}}, \quad \forall j \in B^{\text{WTG}}, \quad \forall t. \\ Q_{j,t}^{\text{WTG}} \leq \bar{Q}_{j,t}^{\text{WTG}}, \quad \forall j \in B^{\text{WTG}} \end{cases} \quad (20)$$

i) *ENS constraints:*

$$\begin{cases} 0 \leq P_{j,t}^{\text{ENS}} \leq k_{j,t}^{\text{cur,P}} P_{j,t}^{\text{Load,PRE}} \\ 0 \leq Q_{j,t}^{\text{ENS}} \leq k_{j,t}^{\text{cur,Q}} P_{j,t}^{\text{Load,PRE}} \end{cases}, \quad \forall t, \forall j \in B^{\text{Load}}. \quad (21)$$

The ENS values are limited to certain ranges related to the load demands.

D. Model Linearization

The above proposed planning model is non-linear due to the existence of the multiplication of three kinds of decision variables, i.e., planning decision variables, scheduling decision variables and information link availability variables. This section first re-expresses those non-linear constraints and then conducts the linearization process for some bi-linear terms by using the big-M method.

1) *Expression (10) Conversion:* Firstly, new variables are introduced: $h_{j,t}^{\text{ESS,charge}}, \rho_{j,l,t}^{\text{ESS,charge}}, h_{j,t}^{\text{ESS,discharge}}, \rho_{j,l,t}^{\text{ESS,discharge}}$. Then, set:

$$\begin{cases} h_{j,t}^{\text{ESS,charge}} = \mu_{j,t}^{\text{charge}} \gamma_{j,t}^{\text{ESS}} \\ h_{j,t}^{\text{ESS,discharge}} = \mu_{j,t}^{\text{discharge}} \gamma_{j,t}^{\text{ESS}} \end{cases} \quad (22)$$

$$\begin{cases} \sum_{l \in \Omega^{\text{ESS}}} P_l^{\text{ESS,charge}} \rho_{j,l,t}^{\text{ESS,charge}} = \sum_{l \in \Omega^{\text{ESS}}} P_l^{\text{ESS}} h_{j,t}^{\text{ESS,charge}} x_{j,l}^{\text{ESS}} \\ \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,charge}} = \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} h_{j,t}^{\text{ESS,charge}} x_{j,l}^{\text{ESS}} \\ \sum_{l \in \Omega^{\text{ESS}}} P_l^{\text{ESS,discharge}} \rho_{j,l,t}^{\text{ESS,discharge}} = \sum_{l \in \Omega^{\text{ESS}}} P_l^{\text{ESS}} h_{j,t}^{\text{ESS,discharge}} x_{j,l}^{\text{ESS}} \\ \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,discharge}} = \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} h_{j,t}^{\text{ESS,discharge}} x_{j,l}^{\text{ESS}} \end{cases} \quad (23)$$

Finally, expression (10) can be replaced by the linear expression (24).

$$\begin{cases} p_{j,t}^{\text{discharge}} \geq \sum_{l \in \Omega^{\text{ESS}}} \underline{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,discharge}} \\ p_{j,t}^{\text{discharge}} \leq \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,discharge}} \\ p_{j,t}^{\text{charge}} \geq \sum_{l \in \Omega^{\text{ESS}}} \underline{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,charge}} \\ p_{j,t}^{\text{charge}} \leq \sum_{l \in \Omega^{\text{ESS}}} \bar{P}_l^{\text{ESS}} \rho_{j,l,t}^{\text{ESS,charge}} \end{cases}, \forall t, \forall j \in B^{\text{ESS}}. \quad (24)$$

2) *Expression (14) Conversion:* New variables are introduced: $h_{j,t}^{\text{CB,IN}}, \rho_{j,t}^{\text{CB,IN}}, \omega_{j,t}^{\text{CB,IN}}, h_{j,t}^{\text{CB,DE}}, \rho_{j,t}^{\text{CB,DE}}, \omega_{j,t}^{\text{CB,DE}}$. Then, set:

$$\begin{cases} h_{j,t}^{\text{CB,IN}} = \delta_{j,t}^{\text{CB,IN}} \gamma_{j,t}^{\text{CB}} \\ h_{j,t}^{\text{CB,DE}} = \delta_{j,t}^{\text{CB,DE}} \gamma_{j,t}^{\text{CB}} \end{cases} \quad (25)$$

$$\begin{cases} \rho_{j,t}^{\text{CB,IN}} = h_{j,t}^{\text{CB,IN}} x_j^{\text{CB}} \\ \rho_{j,t}^{\text{CB,DE}} = h_{j,t}^{\text{CB,DE}} x_j^{\text{CB}} \end{cases} \quad (26)$$

$$\begin{cases} \omega_{j,t}^{\text{CB,IN}} = \rho_{j,t}^{\text{CB,IN}} z_j^{\text{CB}} \\ \omega_{j,t}^{\text{CB,DE}} = \rho_{j,t}^{\text{CB,DE}} z_j^{\text{CB}} \end{cases} \quad (27)$$

Finally, expression (14) can be replaced by the linear expression (28).

$$\begin{cases} y_{j,t}^{\text{CB}} - y_{j,t-1}^{\text{CB}} \leq \omega_{j,t}^{\text{CB,IN}} - \rho_{j,t}^{\text{CB,DE}} \\ y_{j,t}^{\text{CB}} - y_{j,t-1}^{\text{CB}} \geq \rho_{j,t}^{\text{CB,IN}} - \omega_{j,t}^{\text{CB,DE}} \end{cases}, \forall t, \forall j \in B^{\text{CB}}. \quad (28)$$

3) *Expression (16) Conversion:* New variables are introduced: $h_{ij,t}^{\text{Switch,IN}}, \rho_{ij,t}^{\text{Switch,IN}}, h_{ij,t}^{\text{Switch,DE}}, \rho_{ij,t}^{\text{Switch,DE}}$. Then, set:

$$\begin{cases} h_{ij,t}^{\text{Switch,IN}} = \delta_{ij,t}^{\text{Switch,IN}} \gamma_{ij,t}^{\text{Switch}} \\ h_{ij,t}^{\text{Switch,DE}} = \delta_{ij,t}^{\text{Switch,DE}} \gamma_{ij,t}^{\text{Switch}} \end{cases} \quad (29)$$

$$\begin{cases} \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,IN}} = \sum_{l \in \Omega^{\text{Switch}}} h_{ij,t}^{\text{Switch,IN}} x_{ij,l}^{\text{Switch}} \\ \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,DE}} = \sum_{l \in \Omega^{\text{Switch}}} h_{ij,t}^{\text{Switch,DE}} x_{ij,l}^{\text{Switch}} \end{cases} \quad (30)$$

Finally, expression (16) can be replaced by the linear expression (31).

$$\begin{cases} \sigma_{ij,t} - \sigma_{ij,t-1} \leq \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,IN}} - \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,DE}} \\ \sigma_{ij,t} - \sigma_{ij,t-1} \geq \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,IN}} - \sum_{l \in \Omega^{\text{Switch}}} \rho_{ij,l,t}^{\text{Switch,DE}} \end{cases}, \forall t, \forall ij \in E^{\text{Switch}}. \quad (31)$$

4) *Linearization Based on the Big-M Method:* The bilinear terms in (22)-(23), (25)-(27), and (29)-(30) can be converted into linear forms by the big-M method. Taking the bilinear term $h_{j,t}^{\text{ESS,charge}}, x_{j,l}^{\text{ESS}}$ as an example, the linearization process is expressed as follows:

$$\begin{cases} -M \cdot x_{j,l}^{\text{ESS}} \leq \rho_{j,l,t}^{\text{ESS,charge}} \leq M \cdot x_{j,l}^{\text{ESS}} \\ \rho_{j,l,t}^{\text{ESS,charge}} \geq -M \cdot (1 - x_{j,l}^{\text{ESS}}) + h_{j,t}^{\text{ESS,charge}} \\ \rho_{j,l,t}^{\text{ESS,charge}} \leq M \cdot (1 - x_{j,l}^{\text{ESS}}) + h_{j,t}^{\text{ESS,charge}} \end{cases} \quad (32)$$

Obviously, the bilinear term $x_j^{\text{CB}} z_j^{\text{CB}}$ in (2) can be also linearized by the big-M method like expression (32).

IV. ROBUST FORMULATION AND SOLUTION

A robust CPADS planning model is formulated by considering multidimensional uncertainty sets from both cyber space and physical space in this section. And the CCG algorithm is applied to solve the proposed robust planning model.

A. Uncertainty Scenario Sets

1) *Uncertainty From Cyber Space:* As mentioned in Section II, we use FRTU working status to describe information link availability in this study. A series of binary variables (i.e., $\gamma_{j,t}^{\text{ESS}}, \gamma_{j,t}^{\text{CB}}$ and $\gamma_{ij,t}^{\text{Switch}}$) are introduced to characterize the availability of information link. When $\gamma_{j,t}^{\text{ESS}}$ equals 0, it means that an ILF occurs in the ESS FRTU at bus j . The corresponding ESS fails to be actively controlled. On the contrary, it indicates the corresponding information link is working when $\gamma_{j,t}^{\text{ESS}}$ is 1. $\gamma_{j,t}^{\text{CB}}$ and $\gamma_{ij,t}^{\text{Switch}}$ are similar to $\gamma_{j,t}^{\text{ESS}}$. In fact, ILF mainly results from the channel interruption of FRTUs in the interface layer caused by the lack of transmission power, traffic congestion, and external force damage, etc. Thus, in most cases, the occurrence of an ILF on a FRTU can be regarded as an independent event (i.e., the occurrence of an ILF on one FRTU would not affect other FRTUs). Therefore, given the ILF probability distribution of FRTUs, the uncertainty from the cyber space caused by ILF can be expressed in (33).

$$\begin{aligned} & \prod_{t \in T} \prod_{j \in B^{\text{ESS}}} \left[\left(\psi_{j,t}^{\text{ESS,PRES}} \right)^{(1-\gamma_{j,t}^{\text{ESS}})} \cdot \left(1 - \psi_{j,t}^{\text{ESS,PRES}} \right)^{\gamma_{j,t}^{\text{ESS}}} \right] \\ & \cdot \prod_{t \in T} \prod_{j \in B^{\text{CB}}} \left[\left(\psi_{j,t}^{\text{CB,PRES}} \right)^{(1-\gamma_{j,t}^{\text{CB}})} \cdot \left(1 - \psi_{j,t}^{\text{CB,PRES}} \right)^{\gamma_{j,t}^{\text{CB}}} \right] \\ & \cdot \prod_{t \in T} \prod_{ij \in E^{\text{Switch}}} \left[\left(\psi_{ij,t}^{\text{Switch,PRES}} \right)^{(1-\gamma_{ij,t}^{\text{Switch}})} \cdot \left(1 - \psi_{ij,t}^{\text{Switch,PRES}} \right)^{\gamma_{ij,t}^{\text{Switch}}} \right] \\ & \geq \Lambda \end{aligned} \quad (33)$$

where the larger the value of Λ is, the more conservative the decision is. Take logs of both sides in inequality (33), and the linear inequality (34) can be obtained.

$$\begin{aligned} & \sum_{t \in T} \sum_{j \in B^{\text{ESS}}} \left[\log(1 - \psi_{j,t}^{\text{ESS,PRES}}) - \log(\psi_{j,t}^{\text{ESS,PRES}}) \right] \gamma_{j,t}^{\text{ESS}} \\ & + \sum_{t \in T} \sum_{j \in B^{\text{CB}}} \left[\log(1 - \psi_{j,t}^{\text{CB,PRES}}) - \log(\psi_{j,t}^{\text{CB,PRES}}) \right] \gamma_{j,t}^{\text{CB}} \\ & + \sum_{t \in T} \sum_{ij \in E^{\text{Switch}}} \left[\log(1 - \psi_{ij,t}^{\text{Switch,PRES}}) - \log(\psi_{ij,t}^{\text{Switch,PRES}}) \right] \gamma_{ij,t}^{\text{Switch}} \\ & \geq \log(\Lambda) - \sum_{t \in T} \sum_{j \in B^{\text{ESS}}} \log(\psi_{j,t}^{\text{ESS,PRES}}) \\ & - \sum_{t \in T} \sum_{j \in B^{\text{CB}}} \log(\psi_{j,t}^{\text{CB,PRES}}) - \sum_{t \in T} \sum_{ij \in E^{\text{Switch}}} \log(\psi_{ij,t}^{\text{Switch,PRES}}) \end{aligned} \quad (34)$$

In addition, there should be an upper limit on the occurrence number of ILFs for the same FRTU within a full period of operation. This can be expressed as (35).

$$\begin{cases} \sum_{t \in T} \gamma_{j,t}^{\text{ESS}} \geq T - \chi^{\text{ESS}}, & \forall j \in B^{\text{ESS}} \\ \sum_{t \in T} \gamma_{j,t}^{\text{CB}} \geq T - \chi^{\text{CB}}, & \forall j \in B^{\text{CB}} \\ \sum_{t \in T} \gamma_{ij,t}^{\text{Switch}} \geq T - \chi^{\text{Switch}}, & \forall ij \in E^{\text{Switch}} \end{cases} \quad (35)$$

where integer value $\chi^{\text{ESS}}/\chi^{\text{CB}}/\chi^{\text{Switch}}$ are the occurrence maximum number of ILFs for the same ESS/CB/switch FRTU within a full period of operation.

Therefore, the uncertainty from the cyber space can be expressed in the form of (36).

$$\gamma = \left\{ \gamma_{j,t}^{\text{ESS}}, \gamma_{j,t}^{\text{CB}}, \gamma_{ij,t}^{\text{Switch}} \mid \text{expressions(34) - (35)} \right\}, \\ \forall t, \forall j \in B^{\text{ESS}}/B^{\text{CB}}, \forall ij \in E^{\text{Switch}}. \quad (36)$$

In fact, the formula (36) is an N - k uncertainty set considering the probability distribution of ILFs.

2) *Uncertainties From Physical Space*: The uncertainties of wind power output, photovoltaic power output, and load are represented in the form of box-like uncertainty sets, as shown in (37)-(39), respectively.

$$\xi^{\text{WTG}} = \left\{ P_{j,t}^{\text{WTG,PRE}} \mid P_{j,t}^{\text{WTG,PRE}} \leq P_{j,t}^{\text{WTG,PRE}} \leq \bar{P}_{j,t}^{\text{WTG,PRE}} \right\}, \\ \forall t, \forall j \in B^{\text{WTG}} \quad (37)$$

$$\xi^{\text{PVG}} = \left\{ P_{j,t}^{\text{PVG,PRE}} \mid P_{j,t}^{\text{PVG,PRE}} \leq P_{j,t}^{\text{PVG,PRE}} \leq \bar{P}_{j,t}^{\text{PVG,PRE}} \right\}, \\ \forall t, \forall j \in B^{\text{PVG}} \quad (38)$$

$$\xi^{\text{Load}} = \left\{ P_{j,t}^{\text{Load,PRE}}, Q_{j,t}^{\text{Load,PRE}} \mid \begin{pmatrix} P_{j,t}^{\text{Load,PRE}} \leq P_{j,t}^{\text{Load,PRE}} \leq \bar{P}_{j,t}^{\text{Load,PRE}} \\ Q_{j,t}^{\text{Load,PRE}} \leq Q_{j,t}^{\text{Load,PRE}} \leq \bar{Q}_{j,t}^{\text{Load,PRE}} \end{pmatrix} \right\}, \\ \forall t, \forall j \in B^{\text{Load}} \quad (39)$$

Therefore, the uncertainty from the physical space can be expressed in the form of (40).

$$\xi = \left\{ \xi^{\text{WTG}} \xi^{\text{PVG}} \xi^{\text{Load}} \right\}. \quad (40)$$

B. Mathematical Representation

It is mentioned in Section III-A that the proposed bilevel model is traditionally solved iteratively between the upper and lower level. This method is usually time-consuming. In order to improve the calculation efficiency, we transform the iterative process into a single-level problem by uniformly solving the variables in those two levels (the upper investment level and the lower operation level). This single-level model mainly includes two stages when robust uncertainties are considered. In the first stage, the planning decision variables (e.g., $x_{j,t}^{\text{ESS}}, z_j^{\text{CB}}$) and operation decision variables (e.g., $\mu_{j,t}^{\text{charge}}, \delta_{j,t}^{\text{CB,IN}}$) are served as “here-and-now” decisions expressed as μ , which cannot be adjusted until the uncertainties are realized. In the second stage, the rest of discrete variables (e.g., $y_{j,t}^{\text{CB}}, h_{j,t}^{\text{ESS,charge}}, \rho_{j,t}^{\text{ESS,charge}}$) and continuous variables (e.g., $I_{ij,t}, \tilde{V}_{j,t}$) are served as “wait-and-see” decisions (v and y , respectively), which can be adjusted after the first-stage decisions are known and the uncertainties are realized. Consequently, the two-stage robust optimization model is outlined as follows.

$$\min_{\{\mu\}} a^T \mu + \max_{\{\xi\}} \left(\min_{\{y\}\{v\}} (b^T y + c^T \xi) \right) \quad (41)$$

s.t.

$$A\mu \leq d \quad (42)$$

$$By + Cv \leq e \quad (43)$$

$$Dv \leq E\gamma - F\mu \quad (44)$$

$$Gy + Hv = I\xi + f \quad (45)$$

$$\|Jy\|_2 \leq fy \quad (46)$$

Expression (41) consists of the objective functions (2)-(3). Expression (42) denotes the inequality constraints with the first-stage variables, such as expressions (4) and (9). Expression (43) denotes the inequality constraints with only the second-stage variables, such as expressions (24) and (28). Expression (44) denotes the linear constraint by the big-M method in (32), where γ represents a vector composed of binary uncertainty variables (i.e., $\gamma_{j,t}^{\text{ESS}}, \gamma_{j,t}^{\text{CB}}$ and $\gamma_{ij,t}^{\text{Switch}}$). Expression (45) denotes all equality constraints in this paper, where ξ represents a vector composed of uncertainty parameters (e.g., $P_{j,t}^{\text{WTG,PRE}}$). Expression (46) denotes the second order conic (SOC) constraints in (7).

C. Solution Strategy

The original mathematical representation (41)-(46) is divided into a master problem (MP) and a subproblem (SP). The compact form of MP with known worst-case scenarios (i.e., the values of γ^* and ξ^* have been fixed) is given as follows.

$$f^{\text{MP}} = \min_{\{\mu\}} a^T \mu + \eta \quad (47)$$

$$\text{s.t.} \begin{cases} A\mu \leq d \\ By + Cv \leq e \\ Dv + F\mu \leq E\gamma^* \\ Gy + Hv = I\xi^* + f \\ \|Jy\|_2 \leq gy \\ \eta \geq b^T y + c^T \xi^* \end{cases} \quad (48)$$

For the given first-stage variable μ^* in the MP, the SP expressed as (49) and (50) is formulated to obtain the worst-case scenarios.

$$f^{\text{SP}} = \eta(\mu^*) = \max_{\{\xi\}\{y\}} \left(\min_{\{v\}} (b^T y + c^T \xi) \right) \quad (49)$$

$$\text{s.t.} \begin{cases} By + Cv \leq e & (\pi_1) \\ Dv \leq E\gamma - F\mu^* & (\pi_2) \\ Gy + Hv = I\xi + f & (\pi_3) \\ \|Jy\|_2 \leq gy & (\pi_4, \pi_5) \end{cases} \quad (50)$$

SP is a “max-min” problem, which can be converted into a single level “max” problem by dualizing the inner form. However, the uncertainty set in the cyber space and the uncertainty sets in the physical space are quite different in the construction form, making it difficult to obtain the worst-case scenarios from the cyber space and the physical space at the same time. So, it may be a better idea to obtain them separately. Thus, the solution of the dual problem of SP is divided into two steps: step-1 is to solve the dual problem SP-1, while step-2 is to solve the dual problem SP-2. Firstly, after fixing the value of γ^* , SP-1 is solved based on the variable μ^* given by MP to get the worst-case scenario in the physical space (e.g., the value of ξ^*). Then, it is passed to SP-2. Secondly, SP-2 is solved based on both the variable μ^* given by MP and the value given by SP-1 to get the worst-case scenario in the cyber space (e.g., the values of γ^*). Finally, both ξ^* and γ^* are passed to MP.

SP-1 is expressed as (51)-(52).

$$f^{\text{SP},1} = \max_{\{\xi\} \{\pi'_1, \pi'_2, \pi'_3\}} \left(c^T \xi + e^T \pi'_1 + (E\gamma^* - F\mu^*)^T \pi'_2 + (I\xi + f)^T \pi'_3 \right) \quad (51)$$

$$\text{s.t.} \begin{cases} B^T \pi'_1 + G^T \pi'_3 + J^T \pi'_4 + g^T \pi'_5 = b \\ C^T \pi'_1 + D^T \pi'_2 + H^T \pi'_3 = 0 \\ \|\pi'_4\|_2 \leq \pi'_5 \\ \pi'_1, \pi'_2 \leq 0 \\ \pi'_3, \pi'_4 : \text{free variable} \\ \pi'_5 \geq 0 \\ \xi \in (40) \end{cases} \quad (52)$$

The bilinear term $\xi^T I^T \pi'_3$ in (51) can be converted into a linear form by the big-M method, which is explained as follows. Firstly, by introducing a new variable vector κ , (53) can be set.

$$\xi = \xi^{\min} + (\xi^{\max} - \xi^{\min}) \circ \kappa \quad (53)$$

Then, the bilinear term $\xi^T I^T \pi'_3$ can be expressed as (54).

$$\xi^T I^T \pi'_3 = (\xi^{\min})^T I^T \pi'_3 + (\xi^{\max} - \xi^{\min})^T \circ (\kappa^T I^T \pi'_3) \quad (54)$$

For the newly generated bilinear term $\kappa^T I^T \pi'_3$, a new variable vector θ is introduced. By setting $\theta = \kappa^T I^T \pi'_3$, $\kappa^T I^T \pi'_3$ can be linearized via the big-M method, as shown in formula (55).

$$\begin{cases} -M \cdot \kappa^T \leq \theta \leq M \cdot \kappa^T \\ \theta \geq -M \cdot (1 - \kappa^T) + I^T \pi'_3 \\ \theta \leq M \cdot (1 - \kappa^T) + I^T \pi'_3 \end{cases} \quad (55)$$

SP-2 is expressed as (56)-(57):

$$f^{\text{SP},2} = \max_{\{\gamma\} \{\pi''_1, \pi''_2, \pi''_3\}} \left(c^T \xi^* + e^T \pi''_1 + (E\gamma - F\mu^*)^T \pi''_2 + (I\xi^* + f)^T \pi''_3 \right) \quad (56)$$

$$\text{s.t.} \begin{cases} B^T \pi''_1 + G^T \pi''_3 + J^T \pi''_4 + g^T \pi''_5 = b \\ C^T \pi''_1 + D^T \pi''_2 + H^T \pi''_3 = 0 \\ \|\pi''_4\|_2 \leq \pi''_5 \\ \pi''_1, \pi''_2 \leq 0 \\ \pi''_3, \pi''_4 : \text{free variable} \\ \pi''_5 \geq 0 \\ \gamma \in (36) \end{cases} \quad (57)$$

There is also a bilinear term $\gamma^T E^T \pi''_2$ in (56). However, since γ^T belongs to a binary variable matrix, it can be linearized directly by the big-M method in (55).

After solving SP-1 and SP-2, the obtained worst-case scenarios are passed to MP. Then, the robust optimization problem can be solved iteratively by the CCG algorithm [18]. However, expressions (56)-(57) show that SP-2 is still a mixed-integer cone programming problem, so the dual problem only has weak duality, which can result in the fact that a duality gap between the upper and lower bounds is inevitable in the solution process. Thus, an index is defined as shown in (58).

$$\varepsilon = |(UB_s - LB_s) - (UB_{s-1} - LB_{s-1})| \quad (58)$$

Algorithm 1 CCG for Solving the Robust Model

- 1) Set: $LB_0 = 0$, $UB_0 = +\infty$, iteration index $s = 1$.
- 2) Solve MP to get an optimal solution $\{[\mu_s^*], \{v_s^*\}, \{y_s^*\}, \{\eta_s^*\}, \{\gamma_s^*\}$ (i.e., the initial worst-case scenario in the cyber space obtained by solving MP for the first time)] by a mixed integer second-order cone solver, then update the lower bound as $LB_s = \max\{LB_{s-1}, \eta_s^*\}$.
- 3) Fix $\{\mu_s^*\}$ and solve SP.
 - a) Fix $\{\gamma_s^*\}$, solve SP-1 to derive an optimal solution $[\{\xi_s^*\}, \{\pi_{1,s}^*\}, \{\pi_{2,s}^*\}, \{\pi_{3,s}^*\}, \{\pi_{4,s}^*\}, \{\pi_{5,s}^*\}]$.
 - b) Fix $\{\xi_s^*\}$, solve SP-2 to derive an optimal solution $[\{\gamma_s^*\}$ (i.e., the worst-case scenario in the cyber space obtained by solving SP-2)], $\{\pi_{1,s}^*\}, \{\pi_{2,s}^*\}, \{\pi_{3,s}^*\}, \{\pi_{4,s}^*\}, \{\pi_{5,s}^*\}]$ as well as the optimal objective value $f^{\text{SP},2*}$.
 - c) Update the upper bound as $UB_s = \min\{UB_{s-1}, f^{\text{SP},2*}\}$.
- 4) If $\varepsilon \leq \varepsilon_0$, stop and return the optimal solution. Otherwise, update $s = s + 1$, and go to step 5.
- 5) Pass $\{\gamma_s^*\}$ and $\{\xi_s^*\}$ to MP. Solve MP to get the optimal solution $\{[\mu_s^*], \{v_s^*\}, \{y_s^*\}, \{\eta_s^*\}]$ and update the lower bound as $LB_s = \max\{LB_{s-1}, \eta_s^*\}$. Then, go to step 3.

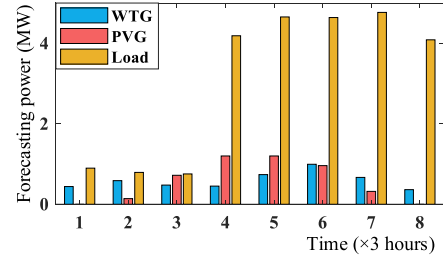


Fig. 2. The forecasting power of total active load and wind/solar unit output.

ε represents the difference between the duality gap in the s -th iteration and the duality gap in the $(s - 1)$ -th iteration, UB_s and LB_s represent the upper bound and the lower bound in the s -th iteration, respectively. The iteration continues until the value of ε is less than the given tolerance ε_0 .

D. Algorithm Implementation

The detailed algorithm implementation procedures are described in Algorithm 1.

V. CASE STUDIES

A modified PG 69 system is used to validate the proposed CPADS planning model in this study. The simulations are implemented by CPLEX 12.8.0 using MATLAB R2018b on a laptop computer with an Intel CPU (2.90GHz) and a 16GB RAM. There are wind turbine generators at buses 37, 53 and 64, and photovoltaic generators at buses 19 and 32. The daily forecasting power curves of the total active load and wind/solar power output per unit during a typical day are shown in Fig. 2. We do not consider multi-stage (or multi-year) load/wind power/solar power data because just a static (single-stage) planning model is proposed in this study. The

TABLE I
ESS CANDIDATE BUSES AND THE RELATED PROBABILITY OF ILFS

Candidate buses	19	32	39	49	53	64
$\psi_{j,t}^{\text{ESS,PRE}}$	25%	15%	15%	25%	25%	15%

TABLE II
CB CANDIDATE BUSES AND THE RELATED PROBABILITY OF ILFS

Candidate buses	39	43	50	53
$\psi_{j,t}^{\text{CB,PRE}}$	15%	15%	25%	25%

TABLE III
SWITCH CANDIDATE BRANCHES AND THE
RELATED PROBABILITY OF ILFS

Candidate branches	6	55	57	69	70	71
$\psi_{j,t}^{\text{Switch,PRE}}$	15%	15%	15%	5%	5%	5%

TABLE IV
CB INVESTMENT PARAMETERS

$Q_j^{\text{CB,step}}$ (Mvar)	c^{CB} (\$10^4\$)	$c^{\text{FRTU,CB}}$ (\$10^4\$)	$\bar{N}^{\text{CB}}$	M^{CB}
0.05	0.3	0.05	8	15

TABLE V
SWITCH INVESTMENT PARAMETERS

Unit types	c_j^{Switch} (\$10^4\$)	$c_j^{\text{FRTU,Switch}}$ (\$10^4\$)	$\bar{N}_i^{\text{Switch}}$
A	0.2	0.05	4
B	0.4	0.05	8

time granularity for short-term operation simulation is 3 hours and thus the whole day time scale is divided into 8 periods. FRTUs at the same bus are assumed to have the same occurrence probability of ILFs in each time period. Meanwhile, In the practical application, not all the locations (or buses) are suitable for installing devices when considering the constraints of natural/cultural environment and other factors. In this study, the candidate buses of ESSs and CBs, the candidate branches of switches, as well as the probability of ILFs of FRTU at each candidate location are shown in Tables I, II and III, respectively. We select these candidate buses just based on this benefit brought to the whole system. We first use the basic load, wind power and solar power data to analysis the overall operation performance (including power loss, wind power curtailment, solar power curtailment, energy not supplied) when installing ESS, CB or switches in those different candidate buses. Finally, we select some suitable candidate locations to verify the effectiveness of the proposed planning method. In fact, this practice that just some buses instead of all the buses/branches are regarded as candidates is common in the current research on active distribution network planning, such as [29]. In the practical application, the ILF probabilities can be obtained via using the method in [16]. Some other technical parameters for invested CBs, switches and ESSs are shown in Tables IV, V and VI, respectively [30]. The discount rate is set as 0.08. The operation equivalent factor is set as 0.5. The value of M is set as 1,000. The value of ε_0 is set as 0.1. Other network data and parameters can be found in [21].

A. Simulation Results

In this section, three uncertainty parameters in the physical space (e.g., τ^{WTG} , τ^{PVG} and τ^{Load}) are introduced for the case studies. Then, (59)-(61) are constructed.

$$\begin{cases} P_{j,t}^{\text{WTG,PRE}} = (1 - \tau^{\text{WTG}})P_{j,t}^{\text{WTG,DET}} \\ \bar{P}_{j,t}^{\text{WTG,PRE}} = (1 + \tau^{\text{WTG}})P_{j,t}^{\text{WTG,DET}}, \forall t, \forall j \in B^{\text{WTG}} \end{cases} \quad (59)$$

$$\begin{cases} P_{j,t}^{\text{PVG,PRE}} = (1 - \tau^{\text{PVG}})P_{j,t}^{\text{PVG,DET}} \\ \bar{P}_{j,t}^{\text{PVG,PRE}} = (1 + \tau^{\text{PVG}})P_{j,t}^{\text{PVG,DET}}, \forall t, \forall j \in B^{\text{PVG}} \end{cases} \quad (60)$$

$$\begin{cases} P_{j,t}^{\text{Load,PRE}} = (1 - \tau^{\text{Load}})P_{j,t}^{\text{Load,DET}} \\ \bar{P}_{j,t}^{\text{Load,PRE}} = (1 + \tau^{\text{Load}})P_{j,t}^{\text{Load,DET}} \\ Q_{j,t}^{\text{Load,PRE}} = (1 - \tau^{\text{Load}})Q_{j,t}^{\text{Load,DET}} \\ \bar{Q}_{j,t}^{\text{Load,PRE}} = (1 + \tau^{\text{Load}})Q_{j,t}^{\text{Load,DET}}, \forall t, \forall j \in B^{\text{Load}} \end{cases} \quad (61)$$

In (59)-(61), $P_{j,t}^{\text{WTG,DET}}$, $P_{j,t}^{\text{PVG,DET}}$, $P_{j,t}^{\text{Load,DET}}$ and $Q_{j,t}^{\text{Load,DET}}$ represent the forecasting data of wind power output, photovoltaic power output, active load, and reactive load, respectively.

In this paper, two different scenarios (i.e., scenario-1 and scenario-2) are selected to reflect the uncertainties from the physical space. For scenario-1, uncertainty parameters are assigned as: $\tau^{\text{WTG}} = 0.2$, $\tau^{\text{PVG}} = 0.2$, $\tau^{\text{Load}} = 0.1$. For scenario-2, uncertainty parameters are assigned as: $\tau^{\text{WTG}} = 0.4$, $\tau^{\text{PVG}} = 0.4$, $\tau^{\text{Load}} = 0.15$. The planning results with different values of Λ in each scenario are shown in Table VII and Table VIII. The location and time period of ILF, as well as the cost loss, are shown in Table IX, where $(B_{\text{ESS}}^{\text{Failure}} - I_{\text{ESS}}^{\text{Failure}})$, $(B_{\text{CB}}^{\text{Failure}} - I_{\text{CB}}^{\text{Failure}})$ and $(E_{\text{Switch}}^{\text{Failure}} - I_{\text{Switch}}^{\text{Failure}})$ represent the location and time period of ILF of ESS, CB and switch, respectively.

By comparing Case 1.x with Case 2.x ($x = 1, 2, 3$, and 4), it can be observed that when the uncertainties in the physical space become larger, the challenges brought by ILFs to the planning and operation of the distribution network becomes greater. Moreover, by comparing the costs in Case 1 (i.e., from Case 1.1 to Case 1.4) and comparing the costs in Case 2 (i.e., from Case 2.1 to Case 2.4), it can be observed that when the uncertainty parameters of the physical space are fixed, both the investment cost and the operation cost have an upward trend with the decrease of Λ . Thus, the total cost increases monotonically.

The change of Λ also results in different planning results. For example, in scenario-1, comparing Case 1.3 with Case 1.2, it can be found that the unit types of the ESS at bus-49 and bus-39 are exchanged, and a switch is added at branch-55. Further, the increase of operation cost is more evident than other costs with the decrease of Λ . This is because the scheduling decision changes after the occurrence of ILF, deviating from the optimal scheduling scheme under normal conditions. Taking the ESS at bus-49 in Case 1.2 as an example, after the ILF of the FRTU occurs in time periods 1 and 2, the ESS that is supposed to be in the charging status cannot be charged. Even though the information link returns to normal from the third time period, the scheduling decision scheme has been changed (see Fig. 3).

Furthermore, it can be observed from Table IX that an ILF occurring in ESS FRTUs (especially the ILF occurring at bus-49 and bus-53 in the ESS FRTU) has a more serious impact on

TABLE VI
ESS INVESTMENT PARAMETERS

Unit types	c_i^{ESS} (\$10 ⁴)	$c_i^{\text{FRTU,ESS}}$ (\$10 ⁴)	P_i^{ESS} (MW)	P_i^{ESS} (MW)	E_i^{ESS} (MWh)	E_i^{ESS} (MWh)	α_j^{charge}	$\alpha_j^{\text{discharge}}$	$\bar{N}_i^{\text{ESS}}$
A	2	0.05	0.2	0	1.5	0	0.9	1.11	5
B	6	0.05	0.5	0	2.5	0	0.9	1.11	8

TABLE VII
PLANNING RESULTS IN SCENARIO-1 ($\tau^{\text{WTG}} = 0.2$, $\tau^{\text{PVG}} = 0.2$, $\tau^{\text{Load}} = 0.1$)

Case	λ	ESS [type]	CB [quantity]	Switch [type]
1.1	5.53×10^{-12}	19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]
1.2	2.05×10^{-13}	19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]
1.3	1.21×10^{-14}	19[B], 32[A], 39[B], 49[A], 53[B], 64[A]	50[15], 53[10]	55[A], 57[A], 71[A]
1.4	3.76×10^{-16}	19[B], 32[A], 39[B], 49[A], 53[B], 64[A]	50[15], 53[10]	55[A], 57[A], 71[A]

TABLE VIII
PLANNING RESULTS IN SCENARIO-2 ($\tau^{\text{WTG}} = 0.4$, $\tau^{\text{PVG}} = 0.4$, $\tau^{\text{Load}} = 0.15$)

Case	λ	ESS [type]	CB [quantity]	Switch [type]
2.1	5.53×10^{-12}	19[B], 32[B], 49[B], 53[B], 64[B]	50[15], 53[13]	55[A], 71[A]
2.2	2.05×10^{-13}	19[B], 32[A], 39[B], 49[B], 53[B]	50[15], 53[13]	55[A], 71[A]
2.3	1.21×10^{-14}	19[B], 32[B], 39[A], 49[B], 53[B], 64[B]	50[15], 53[14]	55[A], 57[A], 71[A]
2.4	3.76×10^{-16}	19[B], 32[B], 39[B], 49[B], 53[B], 64[A]	50[15], 53[15]	55[A], 57[A], 71[A]

TABLE IX
THE LOCATION AND TIME PERIOD OF ILF OCCURRENCE AS WELL AS THE RESULTED COST AND LOSS

Case	$(B_{\text{Failure}}^{\text{ESS}} - t_{\text{Failure}}^{\text{ESS}})$	$(B_{\text{CB}}^{\text{Failure}} - t_{\text{CB}}^{\text{Failure}})$	$(L_{\text{Switch}}^{\text{Failure}} - t_{\text{Switch}}^{\text{Failure}})$	$C^{\text{Inv}} (\$10^4)$	$C^{\text{Ope}} (\$10^4)$	$C^{\text{Total}} (\$10^4)$
1.1	N/A	N/A	N/A	5.6381	3.7265	9.3646
1.2	(49-1), (49-2), (53-1)	N/A	N/A	5.6381	4.3261	9.9642
1.3	(49-1), (49-2), (53-1), (53-2)	N/A	(57-1)	5.6729	4.7274	10.4003
1.4	(49-1), (49-2), (53-1), (53-2), (64-1), (64-2)	N/A	(57-1)	5.6729	4.7419	10.4148
2.1	N/A	N/A	N/A	6.8301	4.0745	10.9046
2.2	(49-1), (49-2), (53-1)	N/A	N/A	6.8301	4.6961	11.5262
2.3	(49-1), (49-2), (53-1), (53-2)	N/A	(55-1)	7.2738	4.7723	12.0461
2.4	(49-1), (49-2), (53-1), (53-2)	(50-4), (50-5)	(55-1)	7.3173	5.2521	12.5694

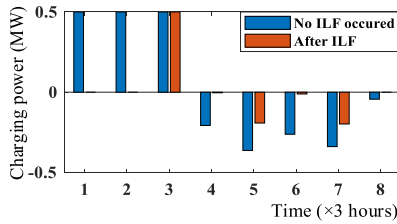


Fig. 3. The scheduling scheme comparison of ESS at bus-49 in Case 1.2 before and after ILF.

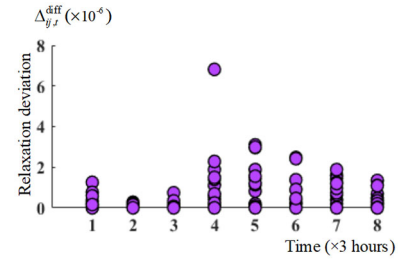


Fig. 4. Relaxation deviation of SOC.

the system operation results compared with both CB FRTUs and switch FRTUs. Furthermore, when the occurrence number of ILFs increases by 2, the increasing rate of the total cost from Case 1.2 to Case 1.3 is 4.377%, while the increasing rate of the total cost from Case 1.3 to Case 1.4 is 0.139%. Similarly, the increasing rate of the cost is 4.511% from Case 2.2 to Case 2.3, and 4.344% from Case 2.3 to Case 2.4. Thus, although the increase in the number of ILFs can lead to an increase in the total cost, the impact brought by ILFs will become gradually smaller. Therefore, the uncertainty in the cyber space should not be too conservatively considered in the practical planning, i.e., it will be acceptable if the uncertainty set can cover the reasonable candidate locations.

B. Analysis on SOC Relaxation

To verify the exactness of the SOC relaxation method, a deviation index $\Delta_{ij,t}^{\text{diff}}$ can be defined as:

$$\Delta_{ij,t}^{\text{diff}} = \left| P_{ij,t}^2 + Q_{ij,t}^2 - \tilde{I}_{ij,t} \cdot \tilde{U}_{j,t} \right| \quad (62)$$

Fig. 4 depicts the relaxation deviation of SOC relaxation in each time period for each branch. Obviously, the deviation of the SOC relaxation gap is sufficiently small, with its order of magnitude being only 10^{-6} , which satisfies the engineering application requirement.

TABLE X
PLANNING RESULTS AS WELL AS THE RESULTING COST AND LOSS WITHOUT CONSIDERING THE UNCERTAINTY FROM CYBER SPACE

Planning Results			Λ	$C^{\text{Inv}} (\$10^4)$	$C^{\text{Ope}} (\$10^4)$	$C^{\text{Total}} (\$10^4)$
ESS [type]	CB [quantity]	Switch [type]				
19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]	5.53×10^{-12}	5.6381	3.7265	9.3646
			2.05×10^{-13}	5.6381	10.2385	15.8766
			1.21×10^{-14}	5.6381	15.2037	20.8418
			3.76×10^{-16}	5.6381	16.2288	21.8668

TABLE XI
ILF OCCURRENCE CIRCUMSTANCES AS WELL AS THE RESULTING COST AND LOSS WHEN THE PROBABILITY DISTRIBUTION OF ILF IS NOT INVOLVED IN THE UNCERTAINTY SET

k	$(B_{\text{ESS}}^{\text{Failure}} - t_{\text{ESS}}^{\text{Failure}})$	$(B_{\text{CB}}^{\text{Failure}} - t_{\text{CB}}^{\text{Failure}})$	$(E_{\text{Switch}}^{\text{Failure}} - t_{\text{Switch}}^{\text{Failure}})$	$C^{\text{Inv}} (\$10^4)$	$C^{\text{Ope}} (\$10^4)$	$C^{\text{Total}} (\$10^4)$
0	N/A	N/A	N/A	5.6381	3.7265	9.3646
3	(49-1), (49-2)	N/A	(71-1)	5.6729	4.4419	10.0448
5	(49-1), (49-2), (32-6)	N/A	(71-1), (71-3)	6.3689	4.9327	11.3016
7	(49-1), (49-2), (32-6), (32-7)	N/A	(57-1), (71-1), (71-3)	6.3689	5.5419	11.9108

C. Comparison Analysis

1) *Regarding the Uncertainty From Cyber Space:* The necessity of the constructed N - k uncertainty set in this paper is verified by the comparison with the results without considering the cyber space uncertainty. All the information link availability variables in the proposed model are set to be 1 (i.e., set $\gamma_{j,t}^{\text{ESS}} = 1$, $\gamma_{j,t}^{\text{CB}} = 1$ and $\gamma_{ij,t}^{\text{Switch}} = 1$). Then, the modified model can be equivalent to a classical ADN planning model. Based on scenario-1, a set of planning results are obtained by simulation calculation using the modified planning model without considering the uncertainty from the cyber space. Then, based on this set of planning results, the resulting cost and loss under different values of Λ are obtained. Details are shown in Table X.

From Table X, it can be observed that the planning results of the method without considering the uncertainty from the cyber space are the same as those in Case 1.1, as shown in Table VII. Meanwhile, the corresponding cost and loss of the method without considering the uncertainty from the cyber space are the same as those in Case 1.1 shown in Table IX when Λ equals 5.53×10^{-12} (i.e., when no ILFs occur). Therefore, when the occurrence of ILFs is not considered, the planning method proposed in this paper is consistent with the existing classical ADNs planning method. However, as the value of Λ decreases (i.e., as the occurrence of ILFs increases), the resulting cost and loss tend to be inconsistent. When Λ respectively equals 2.05×10^{-13} , 1.21×10^{-14} and 3.76×10^{-16} , by comparing Table X with Case 1.2-Case 1.4 in Table IX, it can be observed that although the investment costs of the planning method are the same no matter consider the cyber space uncertainty or not, the operation cost increases greatly with the increasing occurrence of ILFs. Meanwhile, the total costs increase by 69.538%, 122.559% and 133.505%, respectively, which are much higher than the total cost of the proposed method in this paper (i.e., 6.403%, 11.060% and 11.215%, respectively). Thus, the proposed method is demonstrated to be superior in dealing with the uncertainty from the cyber space.

2) *Regarding the Probability Distribution of ILF:* Based on scenario-1, the results for the traditional N - k uncertainty set without considering the probability distribution of ILF is analyzed and compared with those of the proposed method.

The occurrence circumstances of ILF and costs for different values of k are shown in Table XI.

By comparing Table XI with Case 1.1- Case 1.4 in Table IX, it can be observed that when k equals 0, the resulting cost and loss are the same as those in Case 1.1. In other words, when no ILFs occur, the uncertainty set proposed in this paper has the same effect as the traditional N - k uncertainty set. However, as the value of k increases, the ILF occurrence rates of the switch FRTU at branch-71 and the ESS FRTU at bus-32 increase, while the ILF occurrence rate of the ESS FRTU at bus-53 decreases, compared with those in Case 1.1- Case 1.4 in Table IX. This leads to higher investment and operation costs in Table XI than in Case 1.1-Case 1.4 in Table IX. In fact, the probability of ILFs of the switch FRTU at branch-71 is only 5%. The ILF probability of the ESS FRTU at bus-32 is lower than that at bus-53. It is concluded that the N - k uncertainty set constructed in this study is less conservative than that without considering the probability distribution of ILF.

3) *Regarding the Uncertainties From Physical Space:* In order to more effectively analyze what extent uncertainties from physical space the proposed model could withstand, the proposed robust model is compared with the deterministic model corresponding to formulas (41)-(46). Similar to [31], 1,000,000 scenarios are generated using Monto Carlo simulation to select the possible worst-case scenarios related to the physical space. Meanwhile, the ILF occurrence circumstances of the two models are kept consistent, as shown in Table IX. Then, the total cost difference ΔC^{Total} is defined as shown in (63).

$$\Delta C^{\text{Total}} = \frac{C_{\text{DM}}^{\text{Total}} - C_{\text{PM}}^{\text{Total}}}{C_{\text{DM}}^{\text{Total}}} \quad (63)$$

where $C_{\text{PM}}^{\text{Total}}$ and $C_{\text{DM}}^{\text{Total}}$ are the total costs of the proposed robust model and deterministic model under the worst-case scenario related to the physical space, respectively. The total cost difference under different Λ and $[\tau^{\text{WTG}}, \tau^{\text{PVG}}, \tau^{\text{Load}}]$ are shown in Fig. 5.

From Fig. 5, it can be observed that the total cost difference increases with the increase of the uncertainties in the physical space when the value of Λ is fixed. Obviously, the

TABLE XII
PLANNING RESULTS FOR DIFFERENT TEMPORAL RESOLUTIONS

System	Time Granularity	ESS [type]	CB [quantity]	Switch [type]	C^{Inv} (\$10^4\$)	C^{Ope} (\$10^4\$)	C^{Total} (\$10^4\$)	CPU-time
PG69	3h	19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]	5.6381	4.3261	9.9642	<0.5h
	1h	19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]	5.6381	4.2527	9.8908	7.7h
	15min	19[B], 32[A], 39[A], 49[B], 53[B], 64[A]	50[15], 53[10]	57[A], 71[A]	5.6381	4.1228	9.7609	>24h
IEEE33	3h	5[B], 15[B], 24[A]	15[12], 24[9]	5[A], 9[A], 33[A], 35[A]	3.7326	3.1982	6.9308	<0.5h
	1h	5[B], 15[B], 24[A]	15[12], 24[9]	5[A], 9[A], 33[A], 35[A]	3.7326	3.1566	6.8892	5.2h
	15min	5[B], 15[B], 24[A]	15[12], 24[9]	5[A], 9[A], 33[A], 35[A]	3.7326	3.0607	6.7933	>24h

TABLE XIII
COMPUTATIONAL PERFORMANCE OF THE PROPOSED MODEL

Case	Number of iterations	CPU-time (second)
1.1	3	715.283
1.2	4	991.815
1.3	4	1017.977
1.4	4	1123.830
2.1	3	776.379
2.2	4	1001.096
2.3	4	1187.241
2.4	4	1279.403

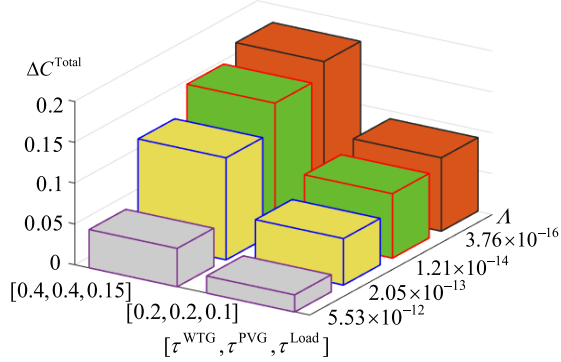


Fig. 5. Total cost difference.

proposed robust model exhibits a better performance under the worst-case scenario related to the physical space.

4) *Regarding the Temporal Resolution:* In order to analyze the differences of planning results for different temporal resolutions, some simulations are carried out based on the PG 69 and an IEEE 33 distribution system with time granularity of 3h, 1h and 15min, respectively. It is noteworthy that the uncertainties from physical space are kept consistent. Meanwhile, the maximum total durations of ILFs in different simulations are consistent, too. The planning results for different temporal resolutions are shown in Table XII.

From Table XII, it can be observed that the planning scheme does not change as the time granularity becomes smaller. In addition, although the operation cost falls as the temporal

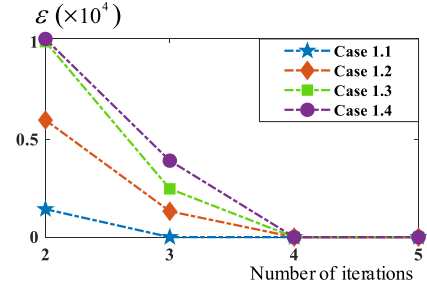


Fig. 6. The value of ε varying with iterations.

resolution becomes higher, the relative cost reduction does not exceed a maximum of 5%. In contrast, the calculation time increases dramatically. Thus, we apply 3-hour time resolution to conduct the integrated planning of cyber-physical active distribution system.

D. Validation of CCG for Solving Robust Optimization Problem

The value of ε varies with the number of iterations as can be seen in Fig. 6. It can be found that when the number of iterations is greater than or equal to 4, the value of ε tends to be 0, i.e., the duality gap hardly changes. Moreover, the optimization results remain unchanged as the number of iterations is greater than or equal to 4. Therefore, it demonstrated that the CCG algorithm is feasible in solving the proposed model in this paper, and the proposed planning method can be practically applied in real scenarios.

E. Computational Performance

The computational performance of the proposed model is shown in Table XII, in terms of iterations and CPU-time. In general, CPU-time depends on the number of iterations. By comparing Case 1.x with Case 2.x ($x = 1, 2, 3$ or 4) in Table XII, it can be observed that when the uncertainties in the physical space become larger, the number of iterations remains unchanged. However, by comparing the CPU-time in Case 1 (i.e., from Case 1.1 to Case 1.4) or in Case 2

(i.e., from Case 2.1 to Case 2.4), it can be observed that the solving difficulty increases since the uncertainty in the cyber space becomes more conservative. Thus, the computational efficiency is mainly affected by the efficiency of seeking the worst-case scenario related to the cyber space.

VI. CONCLUSION

This work proposes a bilevel optimization-based CPADS planning model to optimize the location and selection of active-management elements (e.g., ESSs, CBs and switches), which considers multidimensional uncertainties from both cyber space and physical space. Case studies based on the PG69 distribution system are performed to demonstrate the effectiveness of the proposed method. The main conclusions are drawn as follows.

1) For the uncertainties from both the cyber space and the physical space, the total cost increases with the increase of conservatism degree.

2) The impact of ILF on the ESS FRTUs is the most serious, followed by switch FRTUs and CB FRTUs.

3) Compared with the previous planning methods without considering the uncertainty from the cyber space, the proposed method can reduce a significant amount of unnecessary cost in dealing with ILF.

4) Compared with the classical N - k uncertainty set without the probability distribution of ILFs, the constructed uncertainty set in this study is less conservative and more meaningful.

In the future, we will further study multidimensional uncertainties from both physical and cyber space, such as the conditional-probability distribution formulation for the single-chain EPON in the N - k uncertainty set, FRTU maloperation modelling due to communication failures, or less conservative budget/ellipsoidal uncertainty set integration.

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Hongjun Gao (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Sichuan University, Chengdu, China, in 2011, 2014, and 2017, respectively.

From 2015 to 2016, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of Wisconsin–Milwaukee, Milwaukee, WI, USA. He is currently an Associate Professor with the College of Electrical Engineering, Sichuan University. His research interests include active distribution system planning and operation, economic dispatch, distributed generation integration, and multienergy system optimization.



Xiaodong Lyu received the B.S. degree in electrical engineering from Sichuan University, Chengdu, China, in 2019, where he is currently pursuing the M.S. degree. His research interest is the integrated planning of cyber–physical active distribution system.

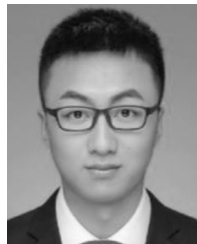


Shuaijia He received the B.S. degree in electrical engineering from Sichuan University, Chengdu, China, in 2018, where he is currently pursuing the Ph.D. degree. His research interests include distribution system planning and multienergy system planning.



Lingfeng Wang (Senior Member, IEEE) received the B.E. degree in measurement and instrumentation from Zhejiang University, Hangzhou, China, in 1997, the M.S. degree in electrical and computer engineering from the National University of Singapore, Singapore, in 2002, and the Ph.D. degree from the Electrical and Computer Engineering Department, Texas A&M University, College Station, TX, USA, in 2008.

He is currently a Professor with the Department of Electrical Engineering and Computer Science, University of Wisconsin–Milwaukee (UWM), Milwaukee, WI, USA. His major research interests include power system reliability, security, and resiliency. He is a recipient of the Outstanding Faculty Research Award of College of Engineering and Applied Science at UWM in 2018. He was an Editor of the IEEE TRANSACTIONS ON SMART GRID, IEEE TRANSACTIONS ON POWER SYSTEMS, and IEEE POWER ENGINEERING LETTERS, and served on the Steering Committee of the IEEE TRANSACTIONS ON CLOUD COMPUTING.



Cheng Wang (Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Tsinghua University, Beijing, China, in 2012 and 2017, respectively.

He was a visiting Ph.D. student with Argonne National Laboratory, Argonne, IL, USA, from 2015 to 2016. He is currently an Associate Professor with North China Electric Power University, Beijing. His research interests include operation and control of integrated energy systems.



Junyong Liu (Member, IEEE) received the Ph.D. degree in electrical engineering from Brunel University, Uxbridge, U.K., in 1998.

He is currently a Professor with the College of Electrical Engineering, Sichuan University, Chengdu, China, where he is the Director with Sichuan Province Key Smart Grid Laboratory. His current research interests include power system planning, operation, and computer applications.